Ministry of Health COVID-19 metrics

Introduction

The New Zealand Te Whatu Ora, Ministry of Health (MoH) reported COVID-19 data metrics to the public on their website:

- https://www.health.govt.nz/covid-19-novel-coronavirus/covid-19-data-and-statistics/covid-19-case-demographics

Data was recorded and updated:

1. Daily between 11 February 2022 and 12 September 2022; and
2. Weekly between 13 September 2022 and 18 February 2024.

The Ministry of Health stopped reporting metrics by vaccination status after 18 February 2024.

Metrics for Cases, Hospitalization, and Deaths of people in New Zealand within 28 days of being reported as a COVID-19 case were stratified on vaccination type and partially by age. This information was collated from the Ministry of Health website and entered into an excel spreadsheet `covid_healthdata.xlsx` available in the GitHub repository. The authenticity of the dataset made available to the public in `covid_healthdata.xlsx` can be verified from snapshots over time still available in the wayback machine.

- https://web.archive.org/web/20260000000000*/https://www.health.govt.nz/covid-19-novel-coronavirus/covid-19-data-and-statistics/covid-19-case-demographics

The random variables the Ministry of Health New Zealand reported to the public were;

Type	Description
Vaccination Uptake_HSU 2021	Numbers of people single/double/triple vaccinated *-not sufficiently stratified by age category*
COVID-19 Cases	Reported cases of COVID-19 by vaccination status
COVID-19 Hospitalizations	Reported cases requiring hospitalization by vaccination status
COVID-19 ICU	Reported cases requiring ICU Hospitalization by vaccination status
COVID-19 Deaths	Reported deaths by vaccination status partially stratified by age category

The information made publicly available by the Ministry of Health represents only a subset of the metrics collected and maintained within its proprietary databases.

The publicly disclosed data lack sufficient age group detail and concise categorisation of vaccination status to support reliable population standardisation (for example, rates per 100,000 people) or comprehensive relative-risk analysis. The publication of Deaths in age categories (0–59, 60–69, 70-79, 80-89, 90+ years) may have been restricted because a small number of deaths within lower age categories could increase the likelihood of identifying individuals and potentially compromising privacy.

Underestimation of the Population

Prior to 4 August 2022, the Ministry of Health underestimated the New Zealand population, this introduced a significant bias in the number of unvaccinated people in New Zealand. This discrepancy arose from its use of the HSU2020 dataset before the HSU2021 dataset was available. This introduced bias into metrics used to estimate relative risks between vaccinated and unvaccinated populations. The following public announcement by the Ministry of Health describes the issue understating its significance to the estimated number of unvaccinated people.

https://www.youtube.com/watch?v=tL06sCjQHxo&t=364s

The underlying issue is that vaccination records are expected to be reasonably accurate as individuals receive vaccinations and are recorded in relevant databases. However, the estimated total population was significantly underestimated. By switching to the updated HSU 2021 database, the residual population classified as unvaccinated increased by an additional 233,000 people. Consequently, the estimated size of the unvaccinated population is substantially larger than had been used in any publication prior to 4 August 2022.

$$\uparrow N_{total} = N_{vaccinated} + \uparrow N_{unvaccinated} \tag{1}$$

This has important implications for interpreting reported infection rates and other relative risk measures in any publication prior to 4 August 2022 when comparing vaccinated to unvaccinated. For example, if prior to using the HSU 2021 database the unvaccinated population was estimated to be 153,000 people, this would increase to $154,000 + 233,000 = 387,000$ unvaccinated, a multiplier of $387/154 = 2.51$ times.

The following screen shot was taken on 8 May 2022 from https://covid19map.co.nz/cases. It illustrates the underestimate in the total population using an estimate of 153,252 for the unvaccinated population.

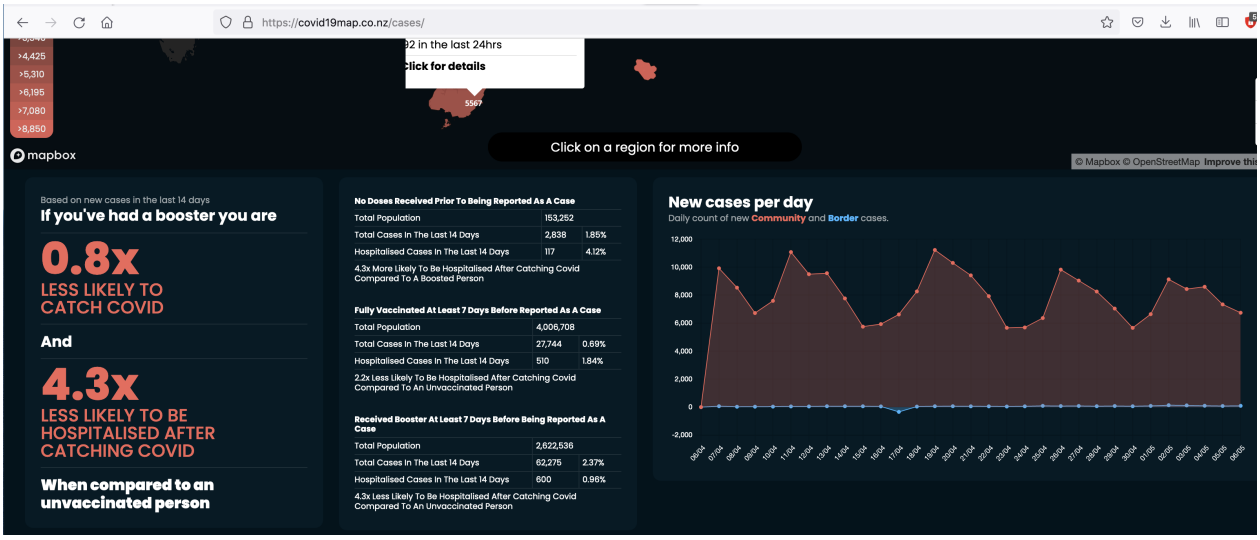


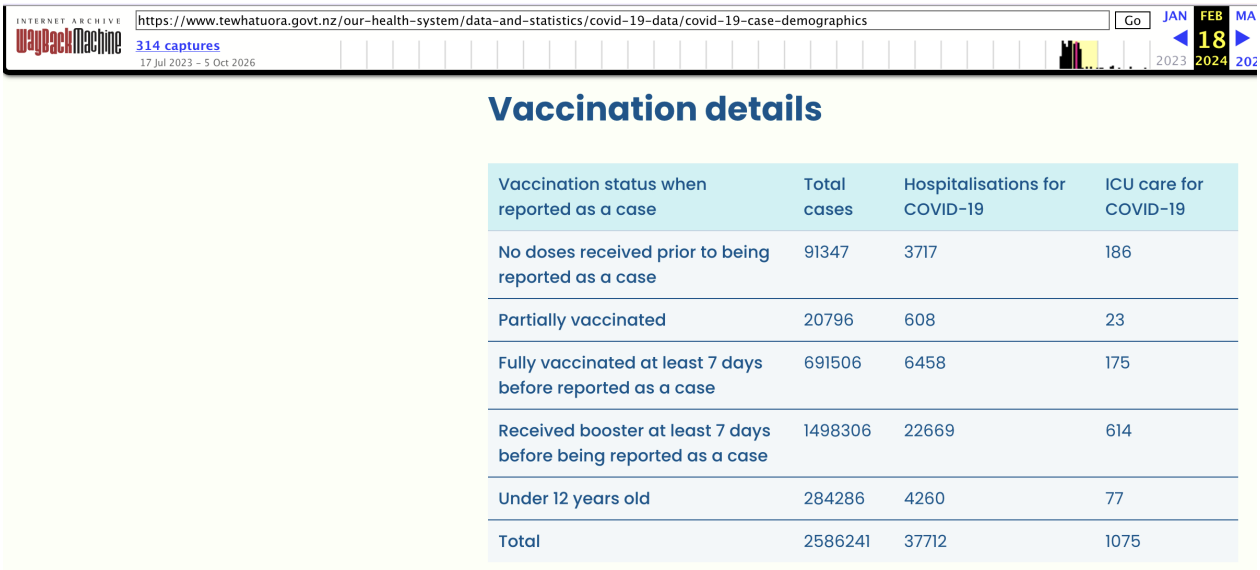
Figure 1: Screenshot of https://covid19map.co.nz/cases on 2022-05-08 at 3.10.02 PM

Note: The relative risk of a person having a booster was 0.8x ***less*** *likely to catch COVID-19 than an unvaccinated person. This is actually in favour of the unvaccinated before using the HSU 2021 population. The equivalent parametrisation is that an unvaccinated person is* $1/0.8 = 1.25x$ ***less*** *likely to catch COVID-19 than a person having a booster.*

$$\frac{2838/153252}{62275/2622536} \approx 0.8x \quad \text{becomes} \quad \frac{2838/(153252+233000)}{62275/2622536} \approx 0.3x \tag{2}$$

Ministry of Health Cases

The data were recorded as cumulative tallies, categorised by vaccination status and updated over time. One advantage of using cumulative data is that discrepancies or inconsistencies in the recorded figures may be identified and corrected by Ministry of Health staff as records are subsequently reviewed and updated. The source of the data came directly from the Ministry of Health dashboard provided to the public.



Vaccination status when reported as a case	Total cases	Hospitalisations for COVID-19	ICU care for COVID-19
No doses received prior to being reported as a case	91347	3717	186
Partially vaccinated	20796	608	23
Fully vaccinated at least 7 days before reported as a case	691506	6458	175
Received booster at least 7 days before being reported as a case	1498306	22669	614
Under 12 years old	284286	4260	77
Total	2586241	37712	1075

Figure 2: Snapshot example of https://web.archive.org/web/20260000000000*/https://www.health.govt.nz/covid-19-novel-coronavirus/covid-19-data-and-statistics/covid-19-case-demographics as at 24 February 2024

The data was collected in the following format:

Date	Status	Cases	Hospitalization	ICU
2022-02-11	Unvaccinated	4443	415	NA
2022-02-11	Single vax	2167	118	NA
2022-02-11	Double vax	4921	63	NA
2022-02-11	Not eligible	3343	53	NA
2022-02-13	Unvaccinated	4498	417	NA
2022-02-13	Single vax	2361	121	NA

Any decrease in a reported value relative to the preceding daily or weekly figure indicates a discrepancy in the published data. Such discrepancies may subsequently be corrected as the Ministry reviews and updates its records.

It should also be noted that the unvaccinated category is subject to a small degree of bias because cumulative data include a number of cases and hospitalisations recorded before vaccines became available. For an objective comparison, counts for all vaccination status categories should ideally be standardised to the same initial observation period. However, this standardisation has not been applied to the observational data presented here.

Reported Cases 2022-02-11 to 2024-02-18

The metric provided was not stratified by any age categories.

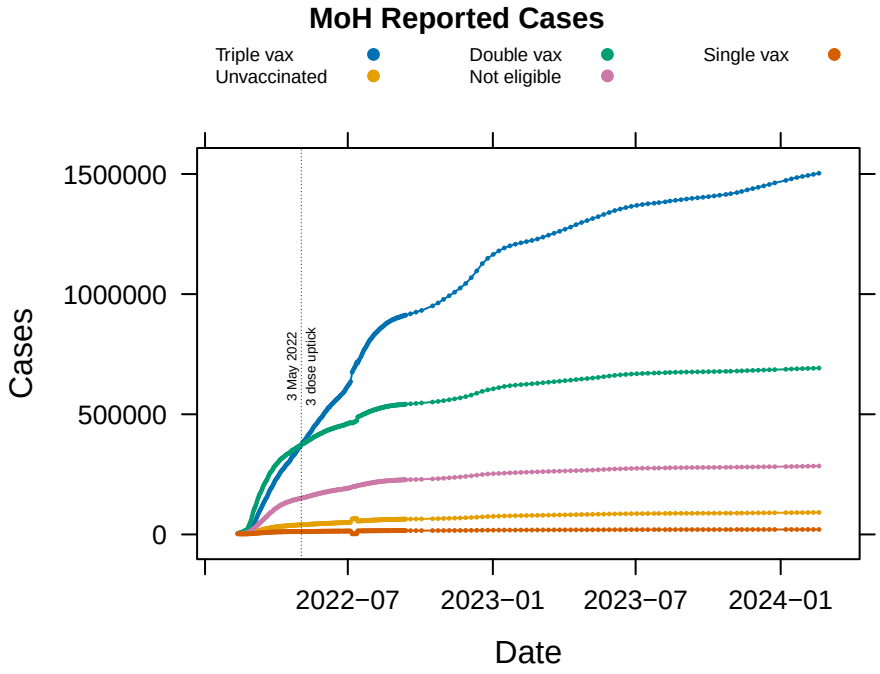


Figure 3: Cases versus Time by Vaccination Status

There is an acceleration of reported COVID-19 cases in the `Triple Vax` category, the crossover point is around 3 May 2022. There were so few cases in the single dose category that a $\log_{10}$ transformation was also used.

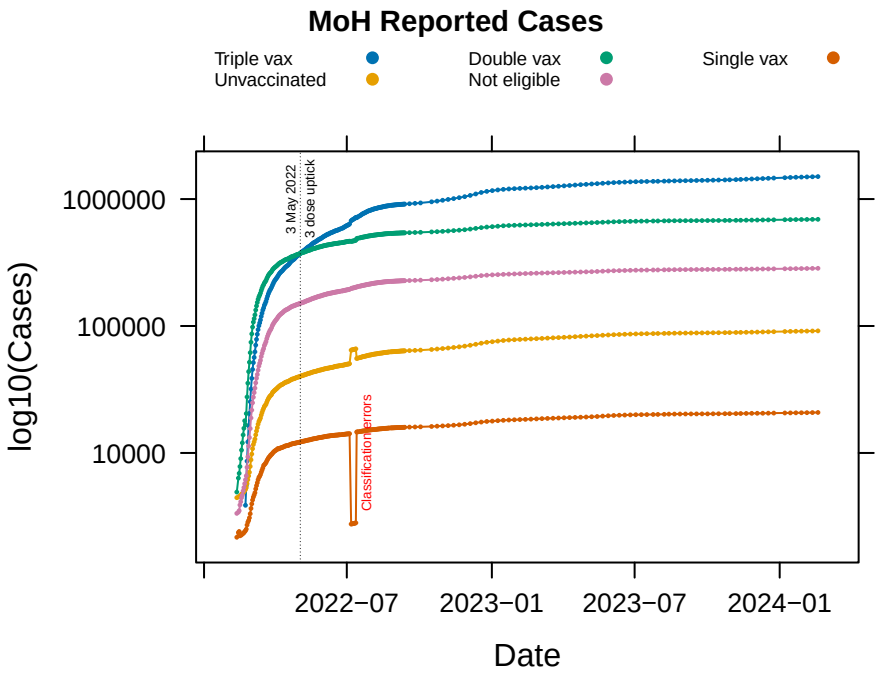


Figure 4: log10(Cases) versus Time by Vaccination Status

This highlights a Ministry of Health classification error in data reporting which is subsequently corrected.

Reported Hospitalizations 2022-02-11 to 2024-02-18

The metric provided was not stratified by any age categories.

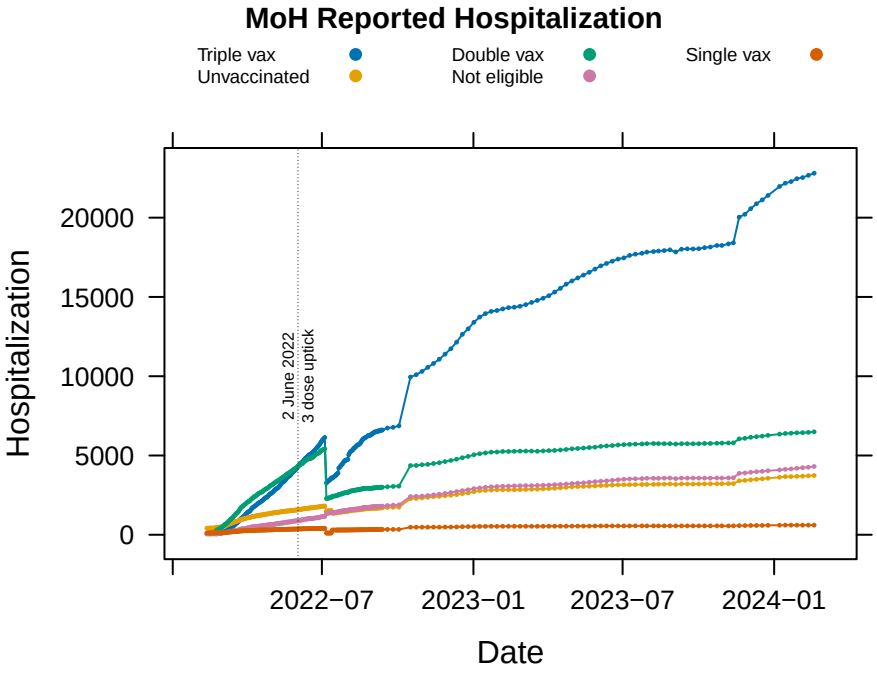


Figure 5: Hospitalization Cases versus Time by Vaccination Status

There is an acceleration of reported COVID-19 cases in the `Triple Vax` category, the crossover point is around 2 June 2922. This is a lag of around a month from the Case crossover point of 3 May 2022.

There are also so few cases in the single dose category that a $\log_{10}$ transformation is required.

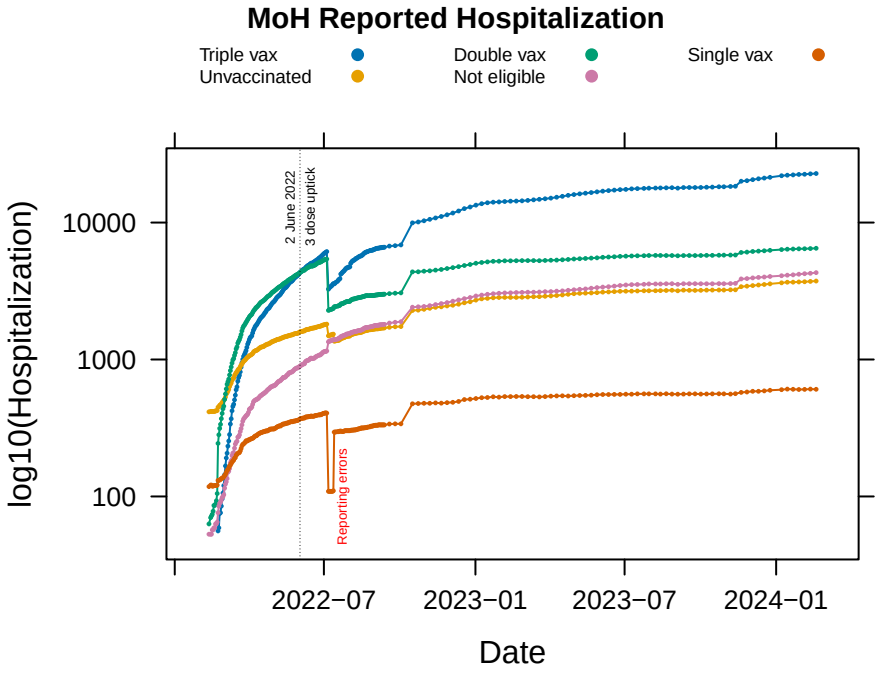


Figure 6: log10(Hospitalization Cases) versus Time by Vaccination Status

This highlights a Ministry of Health reporting error which is subsequently corrected.

Reported ICU 2022-02-11 to 2024-02-18

The metric provided was not stratified by any age categories.

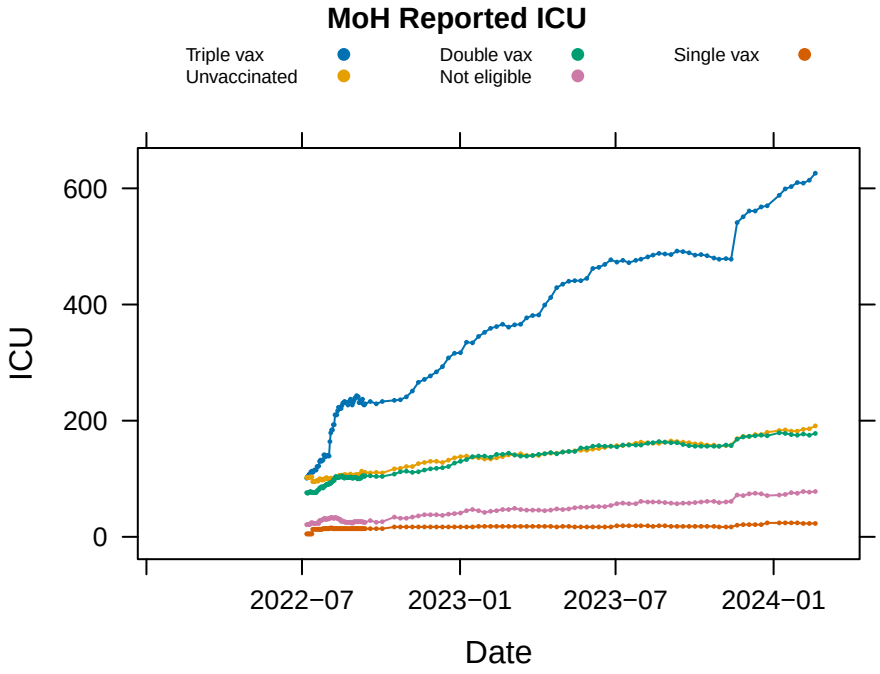


Figure 7: ICU Cases versus Time by Vaccination Status

There are also so few cases in the single dose category that a $\log_{10}$ transformation is required.

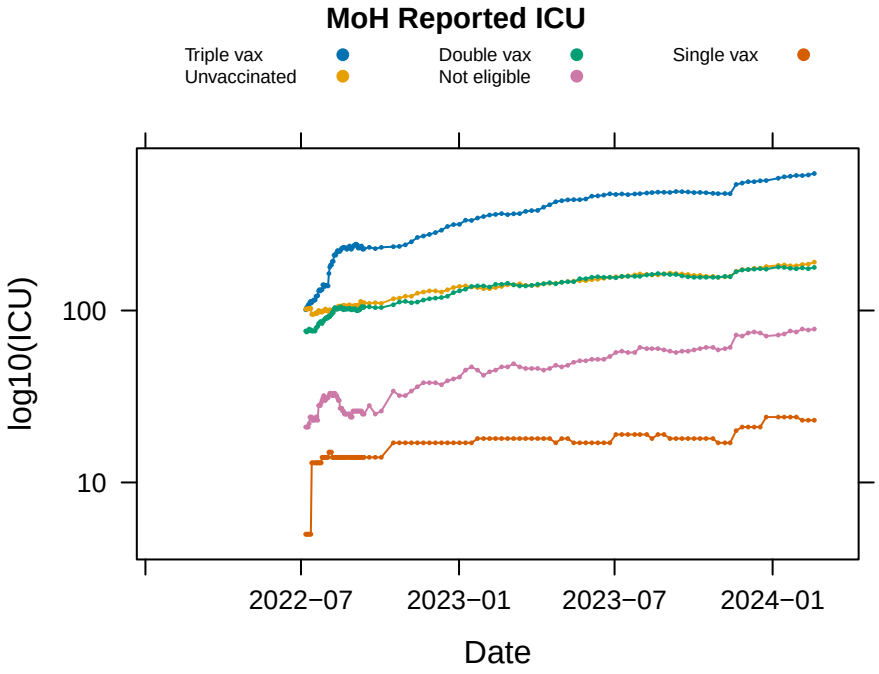
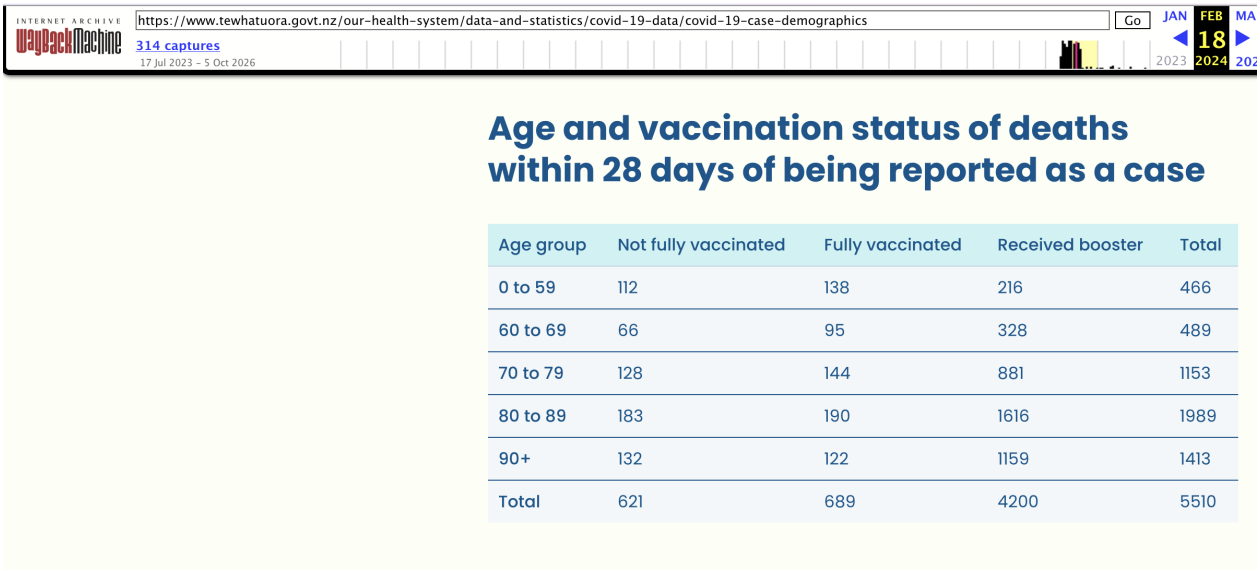


Figure 8: log10(ICU Cases) versus Time by Vaccination Status

Ministry of Health reported Deaths

The data were recorded as cumulative tallies, categorised by vaccination status and (partially) stratified by age over time. One advantage of using cumulative data is that discrepancies or inconsistencies in the recorded figures may be identified and corrected by Ministry of Health staff as records are subsequently reviewed and updated.

The source of the data came directly from the Ministry of Health dashboard provided to the public.



Age group	Not fully vaccinated	Fully vaccinated	Received booster	Total
0 to 59	112	138	216	466
60 to 69	66	95	328	489
70 to 79	128	144	881	1153
80 to 89	183	190	1616	1989
90+	132	122	1159	1413
Total	621	689	4200	5510

Figure 9: Snapshot example of https://web.archive.org/web/20260000000000*/https://www.health.govt.nz/covid-19-novel-coronavirus/covid-19-data-and-statistics/covid-19-case-demographics as at 24 February 2024

The data was collected in the following format:

Date	Age_group	Not_fully_vaccinated	Fully_vaccinated	Received_booster
2022-03-15	0 to 59	17	5	NA
2022-03-15	60 to 69	9	4	3
2022-03-15	70 to 79	14	4	7
2022-03-15	80 to 89	15	14	9
2022-03-15	90+	10	9	5

Any decrease in a reported value relative to the preceding daily or weekly figure indicates a discrepancy in the published data. Such discrepancies may subsequently be corrected as the Ministry reviews and updates its records.

It should also be noted that the unvaccinated category is subject to a small degree of bias because cumulative data include a number of deaths recorded before vaccines became available. For an objective comparison, counts for all vaccination status categories should ideally be standardised to the same initial observation period. However, this standardisation has not been applied to the observational data presented here.

Reported Deaths by Vaccination Status from 2022-02-11 to 2024-02-18

This metric was partially stratified by age category. Essentially, it represents a tally of deaths among people who had contracted COVID-19 within the preceding 28 days. As the vaccination programme initially prioritised older people before progressively extending eligibility to younger age groups, vaccination status is correlated with age. Consequently, age is a potential confounding variable when comparing mortality between vaccination groups. The potential for Simpson's paradox must be considered when interpreting the metrics.

The vaccination status category explanatory variables were reparametrised for plotting:

From	To
Not_fully_vaccinated	$\leq$ 1 dose
Fully_vaccinated	2 doses
Received_booster	$\geq$ 3 doses

Visualising the cumulative COVID-19 Deaths over time conditioned on Vaccination Status, and Age category.

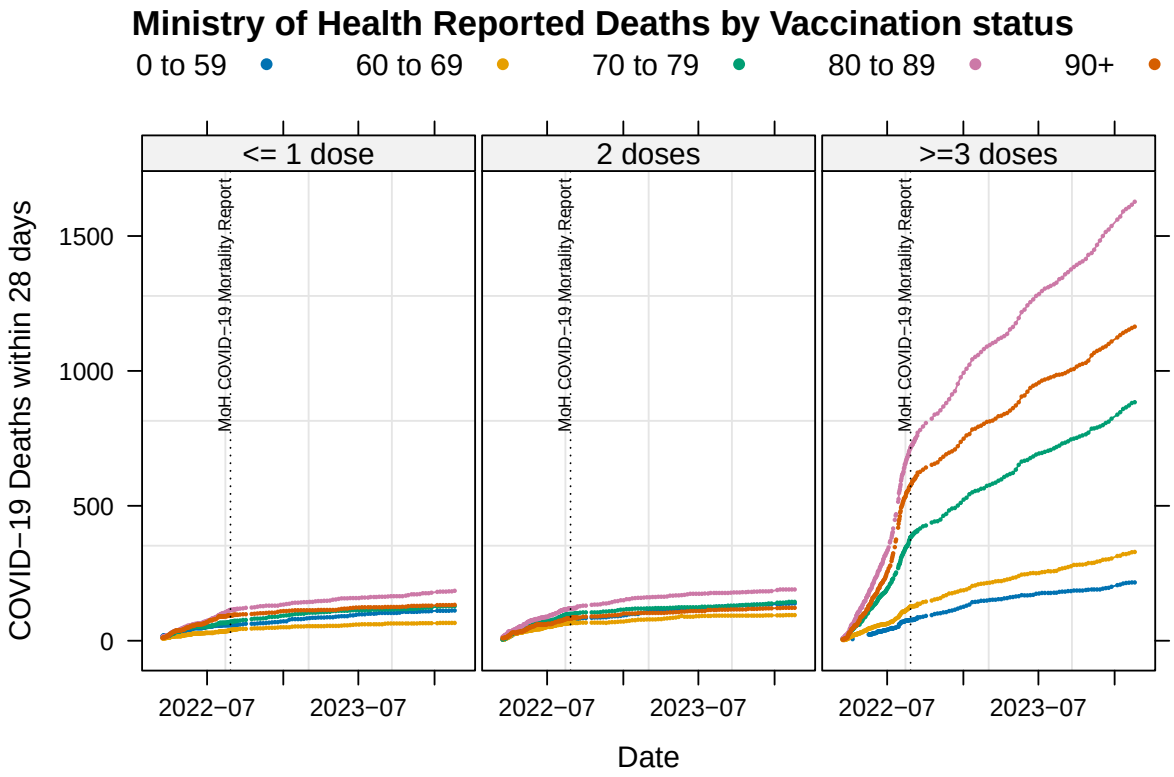


Figure 10: Deaths versus Time conditioned by Vaccination Status and a legend for Age categories. The cumulative trend suggests that deaths due to COVID-19 were significantly higher in the vaccinated group with **>=3 doses** across all age groups.

Vaccination vs Unvaccinated

Insufficient information was provided by the Ministry of Health to estimate relative risks between vaccinated and unvaccinated populations, however the screenshot from 26 November 2023 provides an approximate snapshot at that date of the proportion of the New Zealand population categorized by vaccine status published by the Ministry of Health.

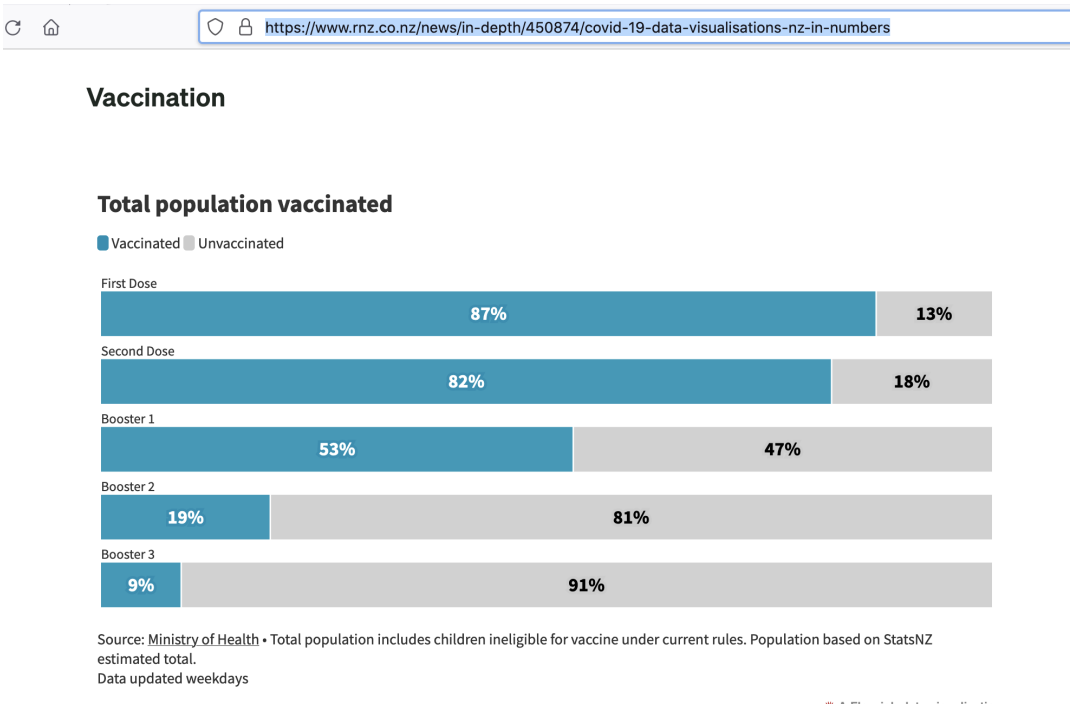


Figure 11: Screenshot from RNZ dashboard on 2023-11-26 at 9.16.30 AM

It can be inferred from Figure 9 that on 26 November 2023 the Vaccination categories in the New Zealand population were:

1. People with ≤ 1 dose equals 18%;
2. Vaccinated people with 2 doses equals $82 - 53 = 29\%$;
3. Vaccinated people with ≥ 3 doses equals 53%.

On the 26 November 2023 when the screenshot capture of the Total Population Vaccinated was taken, the deaths by vaccination status partially stratified by age category was:

Date	Age_group	Not_fully_vaccinated	Fully_vaccinated	Received_booster
2023-11-26	0 to 59	108	134	190
2023-11-26	60 to 69	65	93	295
2023-11-26	70 to 79	121	134	794
2023-11-26	80 to 89	173	183	1483
2023-11-26	90+	129	119	1079

The total deaths by vaccination status were:

Date	Age_group	Not_fully_vaccinated	Fully_vaccinated	Received_booster
2023-11-26	Total	596	663	3841

If the *Received_booster* group represented 53% of the population on 26 November 2023, a naive null hypothesis of no association between booster vaccination status and mortality would suggest that this group would account for approximately 53% of deaths within 28 days of being reported as a COVID-19 case.

However, the reported figure is 75%. This apparent disparity warrants further investigation, even factoring that the *Received_booster* group is likely to be disproportionately represented by older people, who generally have a higher underlying risk of death. Consequently, age is a confounding variable.

$$100 \times \frac{3841}{(596 + 663 + 3841)} = 75.3\% \tag{3}$$

Discussion

On 29 September 2022 the Ministry of Health published the report COVID-19 MORTALITY IN AOTEAROA NEW ZEALAND September 2022.

- https://www.health.govt.nz/system/files/2022-09/covid-19_mortality_in_aotearoa_inequities_in_risk_september_2022_29_sept.v2.pdf

Pages 27 and 28 in the Appendix of this report highlight that the driving variables of *mortality risk* and *case fatality risk* per 100,000 people from 1 January to 26 August 2022 were:

1. Age Group; and
2. Any hospital identified co-morbidity.

It should also be noted that the age weightings used to estimate fully adjusted risk are not provided in the report.

The dashboard data presented above raise questions about why the regression analyses in COVID-19 MORTALITY IN AOTEAROA NEW ZEALAND September 2022 published by the Ministry of Health where they simplified the vaccination status categories, effectively excluding the ≥ 3 doses category from the regression models.

A further question is why, in light of the evidence provided and data available to the Ministry of Health, regression models were not subsequently updated and reported for 2023, 2024 and 2025 using the same statistical methodology. Applying a consistent methodology across these years would have allowed the results to be compared over time.

As with Statistics New Zealand's advice to the Ministry of Health to use the HSU 2021 dataset, independent statistical analysis of publicly released data is recommended to assess the methodology, statistical rigor, and validate the reported findings to evaluate the robustness of conclusions drawn from the Ministry of Healths underlying proprietary collected datasets during COVID-19.

This evidence, drawn from the Ministry of Health's publicly available dashboard data, raises significant questions about the rationale for excluding comparisons of health outcomes between vaccinated and unvaccinated populations from the remit of the *Royal Commission of Inquiry into COVID-19 Lessons Learned (Phase One and Phase Two)*